Ihar Hrachyshka / Senior Software Engineer

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Keywords: Networking, SDN, OVN, Cloud Infra, Linux, Python, C, Golang, Open vSwitch, OpenStack, KubeVirt, Open Source, GenAI, Nix.

> 🔗 At a Glance
>
> **Senior System Software engineer** with extensive **leadership** experience, including **open source** community management. I build **Cloud infrastructure** services, with primary focus on **Networking layer.** I consider **remote** positions only.

🦅 Senior System Software Engineer @ NVIDIA

2025 – Present // Charlotte, NC (Remote)

> 🔗 In Brief
>
> Member of Cloud Engineering, SDN group.

🦅 Principal Software Engineer @ Red Hat, Inc.

2013 – 2025 // Charlotte, NC // Fremont, CA // Brno, Czech Republic

> 🔗 In Brief
>
> **Broad contributions** to Hybrid Cloud portfolio across **multiple products,** including OpenStack Networking (Neutron), OpenShift Virtualization (KubeVirt), OVN (Open vSwitch), and Red Hat AI (InstructLab, Llama Stack).
>
> Proven **leadership** in open-source projects. Currently serving as a **maintainer** for InstructLab, the foundation of RHEL AI's GenAI capabilities. Experienced **speaker** in technical forums.
>
> **Core member** of the OpenStack Networking group. In early 2025, I transitioned to Red Hat AI org where I focus on its **enterprise readiness.**

RHEL AI // 2025 – Present

🖊 Role

Core team member. Upstream maintainer for InstructLab, a LLM fine-tuning toolkit. Miscellaneous contributions in the GenAI orchestration space.

In addition to InstructLab, I currently work with **Llama Stack** community, focusing on its enterprise readiness and integration into future Red Hat products.

OpenStack Networking (Neutron) // 2013 – 2025

🖊 Role

Top core contributor. Trusted engineering voice. Led project teams. (👥)

- Introduced the **network QoS API**. (👥)
- Partial **upgrades** with **no downtime** using a new RPC versioning layer. (👥)
- Fixed **MTU handling** for heterogeneous network types. (talk)
- Served as a **liaison to adjacent projects**, representing OpenStack's interests.

Migrated OpenStack Networking services to **new deployment platform** (Kubernetes, Ansible). (👥) **Network Observability strategy** for OpenStack data plane.

Open vSwitch and OVN // 2019–2022

🖊 Role

OpenStack **liaison** to the project. Delivering **feature parity** with legacy networking solutions, which enabled **complete platform migration** to the new SDN backend. Close **collaboration** with the core team. Building **cross-org** trust.

- VXLAN support enabling encapsulation offload for legacy NICs.
- Stateless ACLs improving **throughput** (**10%+;** more in DPDK environments).
- Reduced **network downtime** for VM live migration from **~10s to ~0.5s** using novel packet cloning approach and dynamic MTU control with ICMP messages. (talk)
- Implemented virtual OVN chassis co-location, **enabling hybrid workloads.**
- Integrated *scapy* into test suite, **improving coverage and correctness.** (talk)
- Multiple functional QinQ enhancements.

KubeVirt // 2018–2019

🖊 Role

Lent my SDN experience to the nascent project. Focused on functional enablement as well as security surface reduction.

- Multi-network connectivity support using Multus CNI.
- Integrated SR-IOV CNI for high-throughput workloads.
- Tightened container capabilities to reduce risk surface.
- Disconnected VMs.

🛜 Lead Software Engineer @ EPAM Systems

2007 – 2012 // Minsk, Belarus

> 🖊 In Brief
>
> I started as a Software Engineer and **quickly advanced** to Senior and Lead roles. Primarily worked for **Infoblox**, becoming an integral part of the core team.

- Contributed to **synchronization layer** for their custom database.
- Implemented key components for DNSSEC zone signing; integration with Active Directory zones; IPv6 support, including developing testing infrastructure.

Also briefly worked on an internal **Linux embedded** multimedia project.

🧯 Software Engineer @ Promwad

2006 – 2007 // Minsk, Belarus

> 🖊 In Brief
>
> **Embedded Linux** engineer working closely with the **hardware team.**

Hardware platform enablement in kernel and distributions. Kernel drivers for custom hardware (graphics, GPIO, ...). Core system services bring-up.

🧪 Quality Engineer @ Velesys

2005 – 2006 // Minsk, Belarus

> 🖊 In Brief
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> Quality Engineer specializing in manual **testing for hardware solutions**, including NAS and kiosks. Wrote test plans and scripts; handled onboarding.

🎓 Education

Minsk Higher State Radio Engineering College

2003 – 2008 // Minsk, Belarus: BSc in Computer Science and Pedagogy.

📝 Other Notes

- **I love my craft.** I know my way with **Unix systems.**
- I'm a **NixOS maintainer** (Darwin team)**.** I know Nix technologies**.**
- **Languages:** English, Belarusian, Russian; limited: Czech, Polish, Ukrainian**.**